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JavaScript Loops

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JavaScript loops **repeat executing a block of code** a number of times.

Loops are fundamental for tasks involving:

- Iteration over values
- Iteration over data structures
- Performing an action multiple times

Loop Type	Description
for	Iterates over values and expressions
while	Iterates over a condition
do...while	Iterates over a condition
for...in	Iterates over the properties of an Object
for...of	Iterates over array like objects
forEach()	Iterates over each element in an Array

The for Loop

The `for` loop is used when the number of iterations is known.

It consists of an initialization (exp1), a condition (exp2), and an increment expression (exp3).

Syntax

exp1 is executed **one time before** the execution of the code block.

exp2 defines **the condition** for executing the code block.

exp3 is executed **every time** a code block has been executed.

Example

```
for (let i = 0; i < 5; i++) {
  text += "The number is " + i + "<br>";
}
```

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The while Loop

The `while` loop executes a block of code as long as a specified condition evaluates to true.

Syntax

```
while (condition) {
  // code block to be executed
}
```

In the following example, the code in the loop will run, over and over again, as long as a variable (i) is less than 10:

Example

```
while (i < 10) {
  text += "The number is " + i;
  i++;
}
```

Note

If you forget to increase the variable used in the condition, the loop will never end. This will crash your browser.

The do...while Loop

The `do...while` loop is similar to the while loop, but guarantees that the code block will be executed at least once, before the condition is checked.

Syntax

```
do {
  // code block to be executed
}
while (condition);
```

The example below uses a `do...while` loop. The loop will always be executed at least once, even if the condition is false, because the code block is executed before the condition is tested:

Example

```
do {
  text += "The number is " + i;
  i++;
}
while (i < 10);
```

Try it Yourself »

Do not forget to increase the variable used in the condition, otherwise the loop will never end!

It is typically used for iterating over object keys.

Syntax

```
for (key in object) {
  // code block to be executed
}
```

A JavaScript `for...in` statement loops through the properties of a person object:

Example

```
const person = {fname:"John", lname:"Doe", age:25};

let text = "";
for (let x in person) {
  text += person[x];
}
```

Try it Yourself »

The for...of Loop

The `for...of` loop iterates over the values of iterable objects such as arrays, strings, Maps, Sets, and NodeLists.

Syntax

```
for (variable of iterable) {
  // code block to be executed
}
```

- **variable**
  For every iteration the value of the next property is assigned to the variable. *Variable* can be declared with `const`, `let`, or `var`.

Other Methods

Array methods like forEach(), map(), filter(), and reduce() provide alternative ways to iterate and manipulate array elements, often offering a more functional and concise approach compared to traditional loops for array-specific operations.

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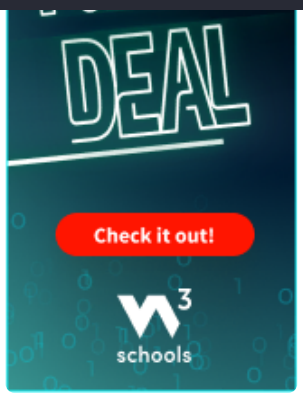

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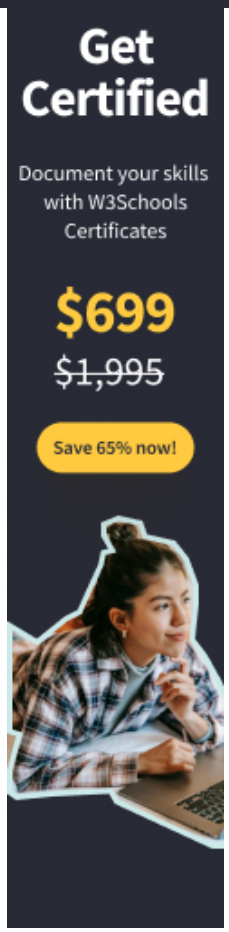

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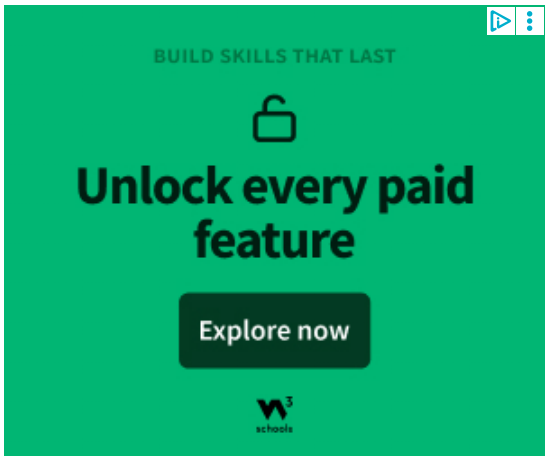

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